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B. Tech Project-II (CE-4291) on

Feasibility Study of short time aerobic composting using vegetables peels

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ABSTRACT

This study investigates the feasibility of short-duration aerobic composting as an efficient method for solid waste management. By analyzing compost prepared from biodegradable municipal solid waste, the research evaluates key parameters such as temperature, moisture content, pH, and nutrient concentrations (NPK) over a 7 to 14-day period. The experiment was conducted using a rectangular pit system to simulate small-scale aerobic composting. Data showed a significant reduction in moisture and increase in organic nutrient content, supporting the viability of rapid composting. The analysis concludes that short-duration composting can be a sustainable, low-cost solution for organic waste treatment, especially in urban and semi-urban settings where land and resources are limited.

KEYWORDS

Aerobic composting, municipal solid waste, short-term composting, organic waste, sustainable waste management, nutrient analysis

1. INTRODUCTION

Municipal Solid Waste (MSW) management has become one of the most pressing urban environmental challenges globally, particularly in developing countries like India. Rapid urbanization, population growth, and changing consumption habits have led to an exponential increase in both the volume and complexity of waste generated [8]. As per recent estimates, urban India alone generates more than 62 million tonnes of solid waste annually, of which nearly 50–60% is biodegradable in nature [6]. The current waste management infrastructure in many Indian cities is under severe stress, lacking the capacity to segregate, collect, and process waste efficiently [4]. A significant fraction of this waste is organic in origin—primarily consisting of food scraps, vegetable peels, and other biodegradable matter from households, eateries, markets, and food processing units [7]. If not treated properly, such waste often ends up in landfills, leading to the production of leachate and the emission of methane (CH_4), a greenhouse gas over 25 times more potent than carbon dioxide in terms of global warming potential [5].

Furthermore, untreated organic waste in open dumps fosters unsanitary conditions, attracts vectors, and risks contaminating nearby water bodies. These environmental and health hazards underscore the need for innovative, decentralized, and sustainable approaches to organic waste treatment. Among the available techniques, aerobic composting is a simple and eco-friendly biological process where aerobic microorganisms decompose organic matter in the presence of oxygen to produce compost a nutrient-rich, humus-like material [1]. However, traditional composting methods can take several weeks to months to reach maturity [2], requiring large space, regular manual turning, and controlled environmental conditions. These requirements make it difficult to implement on a large scale in space-constrained urban institutions.

As a result, recent studies have begun to explore short-duration aerobic composting techniques, which aim to complete the process within a much shorter timeframe typically two weeks without compromising the quality of compost. This study is situated within this context and investigates the feasibility of implementing a small-scale, rapid composting system using vegetable peels and locally available additives. By focusing on decentralized waste treatment models suitable for campuses, communities, and individual households, this approach aligns with the broader sustainability goals outlined in the United Nations Sustainable Development Goals (SDGs) related to climate action, responsible consumption, and sustainable cities.

2. OBJECTIVE AND SCOPE

2.1. Objective

- To assess the feasibility of short-time aerobic composting using vegetable peels.
- To complete the composting process within a one or two week period.
- To monitor key parameters: temperature, pH, moisture content, C/N ratio, TKN, COD, and $\text{NH}_4^+\text{-N}$.
- To evaluate compost quality and maturity using simple, low-cost materials.

2.2. Scope

- Focus on decentralized composting suitable for urban areas, institutions, and households.
- Use of easily available materials like cow dung, compost seed, and microbial inoculants.
- Promote waste reduction, soil nutrient recycling, and low environmental impact.

- Align the method with Sustainable Development Goals (SDGs) on waste management and climate action.

3. MATERIALS AND METHODS

3.1. MATERIALS

3.1.1. VEGETABLES PEELS

Vegetable peels were collected from two sources within the IEST Shibpur campus: Richardson Hall and the Institute Canteen. The collected organic waste primarily included peels of potato, carrot, spinach, broccoli, tomato, lettuce, cucumber, onion, and eggplant. After collection, the peels were manually chopped into smaller pieces to ensure uniformity and facilitate efficient microbial decomposition.



Figure 1: Vegetables Peels collected from household waste

3.1.2. COMPOST SEED

For Batch 1, Bhudan Compost—sourced from the Dhapa Dumping Yard, Kolkata—was used as the initial compost seed. The same compost was used again in Batch 2. From Batch 3 to Batch 6, compost from the immediately preceding

batch was used as the seed material. This continuous inoculation helped maintain microbial consistency and improved composting efficiency.



Figure 2: Dhapa Composting Yard, Kolkata



Figure 3: Budhan Compost and Compost collect from batch

3.1.3. INOCULUM

In Batch 2, the TrustBasket Bokashi Compost Maker was used as a microbial inoculum to enhance decomposition. For Batch 6, a homemade inoculum was prepared by mixing 250 grams of compost from a previous batch, 50 grams of curd (as a source of lactic acid bacteria), and 800 grams of water. This inoculum mixture was used to further stimulate microbial activity in the compost pile.

3.1.4. COW DUNG

Raw cow dung was collected from a local cow farm and used in Batch 3 as a nitrogen-rich organic material and microbial booster. In Batch 6, dried dung cakes were utilized, which were prepared using a traditional mixture of cow dung, cow urine, and husk. These served as natural compost activators.

3.1.5. ANALYTICAL AND MEASURING INSTRUMENTS

A 3-in-1 Soil Tester was used to monitor key composting parameters such as temperature, moisture content, and pH. Regular monitoring ensured the composting environment remained optimal for microbial activity and organic matter breakdown.

A Delta digital weighing machine was used to record the weight of raw materials and final compost output, ensuring accurate mass tracking across all batches.

Total Kjeldahl Nitrogen (TKN) and Ammonium Nitrogen ($\text{NH}_4^+\text{-N}$) were measured using a Thermo Scientific Ionometer, enabling precise detection of nitrogen species in the compost samples.

Chemical Oxygen Demand (COD) analysis was carried out using the Spectroquant TR 320, a photometric digestion system suitable for high-accuracy organic matter quantification in compost leachate or extracts.



Figure 6: Soil Tester and Ionmeter



Figure 7: COD testing Machine (Spectroquant TR 320)

3.1.6. GERMINATION SEED MIX

For assessing compost maturity and plant growth potential, **TrustBasket Seed Germination Mix** was used. The germination mix was blended with the extracted compost from each batch and used for seed germination tests to evaluate the phytotoxicity and nutrient effectiveness of the compost.

3.2. METOHDS

3.2.1. FABRICATION OF REACTOR

The fabrication of the composting reactor was carried out with a focus on cost-effectiveness, durability, and operational simplicity. The primary objective was to design a unit capable of efficiently processing biodegradable waste under controlled aerobic conditions. The reactor was constructed using readily available materials and involved both mechanical assembly and structural modifications.

Material Selection

Main Body: A cylindrical PVC bucket (volume: approx. 15-20 liters) was selected due to its lightweight nature, corrosion resistance, and ease of drilling for aeration holes.

Leachate Collection System: A small outlet was provided at the bottom of the reactor for leachate collection.

Construction Steps

Preparation of Outer Bucket:

Holes ($\varnothing$ 8–10 mm) were drilled uniformly around the body to allow air circulation. The bottom portion was slightly elevated and a sloped plate was placed to direct leachate to the outlet.

Fabrication of Inner Drum:

Multiple perforations ($\varnothing$ 5–10 mm) were made to allow passive aeration. The drum was welded and riveted for rigidity and placed concentrically inside the outer bucket.

Frame and Support:

A plastic frame was welded using + shapes to elevate the bucket. Rubber bushes or plastic washers were used to isolate vibrations and prevent corrosion at contact points.

Lid and Insulation:

A tight-fitting plastic lid was used to cover the bucket, ensuring minimal odor release.

Final Assembly:

All components were assembled and checked for stability. The leachate outlet was connected to a small bottle or container for collection.

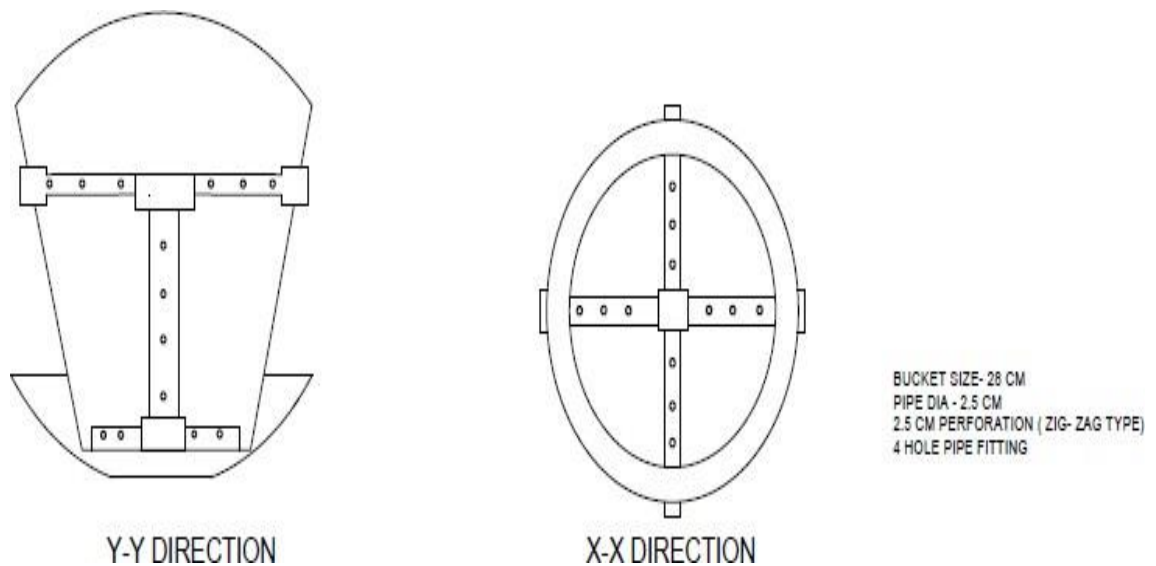




Figure 9: Composting Reactor

3.2.2. COLLECTION OF VEGETABLES PEELS

Vegetable peels were collected regularly from two primary sources within the IEST Shibpur campus: **Richardson Hall** and the **Institute Canteen**. The collected waste primarily included peels and trimmings of **potato, carrot, spinach, broccoli, tomato, lettuce, cucumber, onion, and eggplant**.

The collection was done manually in clean, labeled containers to avoid cross-contamination with non-biodegradable waste. After transportation to the composting site, the peels were chopped into small pieces to increase the surface area and enhance microbial degradation during composting.

3.2.3. PROCUREMENT OF COMPOST

For initiating the composting process, compost seed material was procured in two phases. In **Batch 1**, compost known as *Bhudan Compost* was collected from the **Dhapa Dumping Yard, Kolkata**, and used as the initial seed to introduce

beneficial microorganisms into the system. The same compost was again used for **Batch 2** to maintain microbial activity.

From **Batch 3 onwards**, compost from the previous batch was used as the seed material for the next, ensuring a continuous and consistent microbial population. This method of **successive batch inoculation** helped in sustaining microbial efficiency and reducing dependency on externally sourced compost.

3.2.4. MIXING RATIO VEGETABLE WASTE WITH COMPOST, INOCULUM, COW DUNG

The mixing process was carefully designed to ensure homogeneous blending of all input materials—vegetable waste, compost seed, and inoculum—before composting. For each batch, **pre-weighed quantities** of chopped vegetable peels (collected from Richardson Hall and the Institute Canteen), compost seed (starting from Bhudan Compost in Batch 1 and reused from previous batches thereafter), and microbial inoculum were added according to specific ratios.

In **Batch 2**, *TrustBasket Bokashi Compost Maker* was used as the inoculum, while in **Batch 6**, a natural inoculum made from **250 g compost, 50 g curd, and 800 ml water** was prepared and applied. In **later batches**, cow dung (both raw and in cake form) was also incorporated as a microbial and nitrogen source.

Mixing Ratios Per Batch

Table 1: MIX RATIO OF VEGETABLES WITH COMPOST, INOCULUM, COW DUNG

Batch	Reactor No.	Vegetable Peels (kg)	Compost Seed (kg)	Inoculum (kg)	Cow Dung (kg)	Mixing Ratio (Veg:Compost:Inoculum/Cow Dung)
1	1	1	0.5	-	-	1:0.5
	2	2	1	-	-	2:1
	3	3	1.5	-	-	2:1
2	1	1	0.35	0.15	-	1:0.35:0.15
	2	2	0.75	0.3	-	2:0.75:0.3

Batch	Reactor No.	Vegetable Peels (kg)	Compost Seed (kg)	Inoculum (kg)	Cow Dung (kg)	Mixing Ratio (Veg:Compost:Inoculum/Cow Dung)
	3	3	1.05	0.45	-	3:1.05:0.45
3	1	1	1	-	0.1	1:1:0.1 (Cow Dung)
	2	2	2	-	0.2	2:2:0.2 (Cow Dung)
	3	0.55	0.55	-	0.05	0.55:0.55:0.05 (Cow Dung)
4	1	1	1	-	0.04	1:1:0.04 (Cow Dung)
	2	1.5	1.5	-	0.05	1.5:1.5:0.05 (Cow Dung)
	3	0.53	0.53	-	0.02	0.53:0.53:0.02 (Cow Dung)
5	1	1	1	-	0.09	1:1:0.09 (Cow Dung)
	2	2	2	-	0.18	2:2:0.18 (Cow Dung)
	3	1	1	-	-	1:1
6	1	1	1	(Inoculum mixed)	-	1:1 + 250g compost + 50g curd + 800ml water (mix)
	2	1	1	-	0.5	1:1:0.5 (Cow Dung)
	3	1	1	0.6	0.5	1:1:0.6 (Inoculum) + Cow Dung

3.3. REGULAR MONITORING OF PARAMETERS (pH, MOISTURE CONTENT, TEMPERATURE, C/N RATIO, COD, TKN, NH₄⁺_N)

3.3.1. Procedure for TKN (Total Kjeldahl Nitrogen):

Sample Preparation:

Collect 50 g of sample from reactors. Mix with 100 mL of distilled water and boil using a heater. Filter the boiled sample using a pre-weighed filter paper. Dry the filter paper in an oven at 105°C for 2 hours. Record the final dry weight of the filter paper. The difference in weight gives the solid residue (used later for conversion to mg/g).



Figure 10: Sample (Collected from reactor after decompose) Preparation

Digestion:

Dilute the sample 10 times with distilled water. Mix 50 mL of the diluted sample with 50 mL of digestion reagent (K_2SO_4 , CuSO_4 , H_2SO_4). Digest the mixture at 60–70°C for 20 minutes.



Figure 11: Digestion

Measurement (Ionmeter with Double Junction Electrode):

Calibrate the ionmeter using $(\text{NH}_4)_6\text{Cl}$ standards at 100 mg/L, 10 mg/L, and 1 mg/L concentrations. Use a 10:1 ratio of standard to ISA (Ionic Strength Adjuster) during calibration. Add 1 mL of CH_3COOH to each standard during calibration. Prepare the digested sample by adding 1 mL ISA. Measure the TKN concentration using the ionmeter.

Data Conversion:

Ionmeter readings are obtained in mg/L. Convert values to mg/g by dividing by the weight of the dried filter residue.

3.3.2. Procedure for Inorganic $\text{NH}_4^+\text{-N}$ (Ammonium Nitrogen):

Sample Preparation:

Use the same diluted sample (10× dilution). Add 1 mL of ISA to the sample.

Measurement (Ionmeter with Double Junction Electrode):

Follow the same calibration steps as for TKN. Measure the $\text{NH}_4^+\text{-N}$ concentration using the ionmeter.

Data Conversion:

Convert the ionmeter reading from mg/L to mg/g.

3.3.3. Procedure for COD (Chemical Oxygen Demand):

Sample Preparation:

Follow the same initial preparation steps as described in the TKN procedure.

Dilute the sample with distilled water to ensure that it **does not turn green** upon the addition of concentrated H_2SO_4 (this indicates appropriate dilution and proper reaction conditions).

Digestion Mixture:

Take **2.5 mL of the diluted sample**. Add **1.5 mL of $\text{K}_2\text{Cr}_2\text{O}_7$** (Potassium Dichromate). Add **3.5 mL of concentrated H_2SO_4** slowly and carefully.

Blank Preparation:

Cold Blank: 1.5 mL of $\text{K}_2\text{Cr}_2\text{O}_7$ + 3.5 mL of H_2SO_4 (no heating).

Hot Blank: Same as cold blank but digested with sample.

Digestion:

Digest the **sample and hot blank** at **148°C for 2 hours** using **Spectroquant TR 320** instrument. After digestion, cool both the sample and hot blank to room temperature.

Titration:

Titrate **sample, hot blank, and cold blank** using **0.01 M FAS (Ferrous Ammonium Sulfate)**. Record the burette readings for all three.

Calculation of COD:

$$\text{COD (mg/L)} = \frac{(V_{\text{blank}} - V_{\text{sample}}) \times M \times 8 \times 1000 \times D.F.}{V_{\text{sample}}}$$

Where:

- V_{blank} = Burette reading of the blank (hot)
- V_{sample} = Burette reading of the sample
- M = Molarity of FAS (0.01 M)
- 8 = Equivalent weight of O_2
- $D.F.$ = Dilution factor
- V_{sample} = Volume of the sample taken (2.5 mL)

Conversion to mg/g:

- Convert COD value from mg/L to **mg/g** using:

$$\text{COD (mg/g)} = \frac{\text{COD (mg/L)}}{\text{Solid residue (g)}}$$

Solid residue is obtained from the dried filter paper used in the sample preparation.

3.3.4. C/N Ratio Procedure:

The **C/N ratio** is calculated as the ratio of **Carbon (C)** to **Nitrogen (N)** in the sample. In wastewater and environmental analysis, this is commonly represented as:

$$\text{C/N Ratio} = \frac{\text{COD (mg/g)}}{\text{TKN (mg/g)}}$$

Where:

- **COD (mg/g)** represents the organic carbon content indirectly (as Oxygen Demand).
- **TKN (mg/g)** represents the total nitrogen content.

3.3.5. Moisture Content Determination Procedure:

Sample Preparation:

Weigh **10 g** of the wet sample collected from the reactor. Place the sample into a clean, pre-weighed **crucible**.

Drying:

Dry the sample in a hot air oven at **105°C** for **2 hours**.

Post-Drying Measurement:

Remove the crucible from the oven, and weigh the **dried sample**.

Calculation:

$$\text{Moisture Content (\%)} = \frac{\text{Initial weight} - \text{Dried weight}}{\text{Initial weight}} \times 100$$

3.3.6. Temperature and pH Measurement Procedure:

Instrument Used:

Use a **Soil Testing Thermometer** that includes both **temperature** and **pH** measurement probes.

Procedure:

Insert the probe of the soil testing thermometer directly into the **wet reactor sample**. Wait for the readings to stabilize.

Readings:

- Record the **temperature** in °C.

- Record the **pH** as shown on the instrument.



Figure 14: Soil Tester Machine

3.3.7. Preparation of Final Compost:

After the composting period, each batch was monitored until key parameters stabilized, including temperature, pH, moisture content, TKN, COD, C/N ratio, and $\text{NH}_4^+\text{-N}$ levels, which indicated the maturity of the compost. Once the material was stabilized, it was prepared for refinement.

In Batch 1, after one month of composting, the entire composted mass was stored for curing and then subjected to sieving using a 1.18 mm mesh on 26 February 2025. Before sieving, TrustBasket Seed Germination Mix was added and blended with the composted mass to help regulate moisture content and improve the physical texture of the final product. This step ensured smoother sieving and enhanced the quality of the final compost.



Figure 15: Adding Germination Seed Mix at Batch-1

The sieving process separated the fine, well-decomposed compost, which was collected as the final product, from the coarser residues. The fine compost was not only suitable for soil application and seed germination testing but was also retained as compost seed for Batch 3 and subsequent batches. This practice of reusing mature compost as seed ensured microbial continuity, minimized external input, and maintained a sustainable and efficient batch-wise composting cycle.



Figure 16: Sieving and The Final Product

4. RESULTS AND DISCUSSION

4.1. pH Variations

REACTOR-1 (All Batches)

The pH levels in REACTOR-1 were measured across six distinct batches. The average pH values varied slightly among the batches, ranging from **5.24 (Batch-3)** to **6.11 (Batch-5)**. The lowest observed pH was **4.7** (in both Batch-3 and Batch-6), indicating more acidic conditions, while the highest was **7.2** in Batch-5, approaching neutral. Batch-2 maintained a consistently near-neutral environment with a narrow fluctuation (min: 5.7, max: 6.3). In contrast, Batch-5 showed the widest pH range (5.4–7.2), suggesting significant biological or chemical activity. These variations across the reactor batches highlight differences in metabolic activity, substrate consumption, or microbial dynamics.

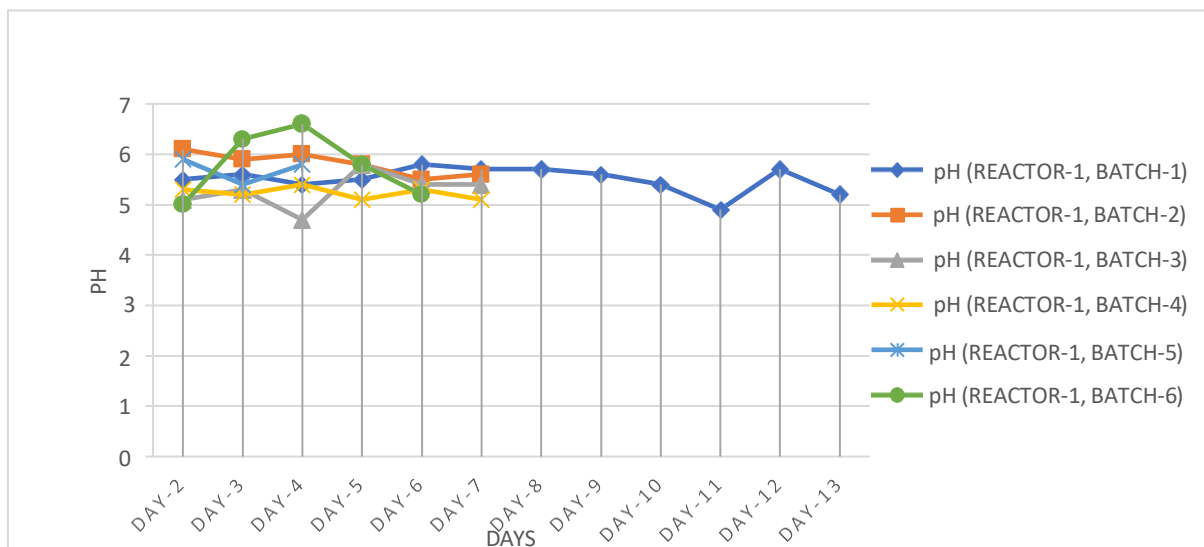


Figure 17: pH variations in Reactor-1

REACTOR-2 (All Batches)

The pH readings in REACTOR-2 showed a relatively broader range of fluctuation compared to REACTOR-1, with values spanning from **4.7 to 7.3**. Batch-5 started

with a notably high pH of **7.2** on Day-2, which gradually declined, indicating initial alkaline conditions that stabilized over time. Batch-4 exhibited a temporary peak reaching nearly **6.8**, while Batch-3 remained more acidic with a low of **4.7**. Batch-1 maintained a moderate and consistent trend similar to REACTOR-1. The largest fluctuations were observed in Batch-2 and Batch-5, possibly due to varying microbial activity or substrate dynamics.

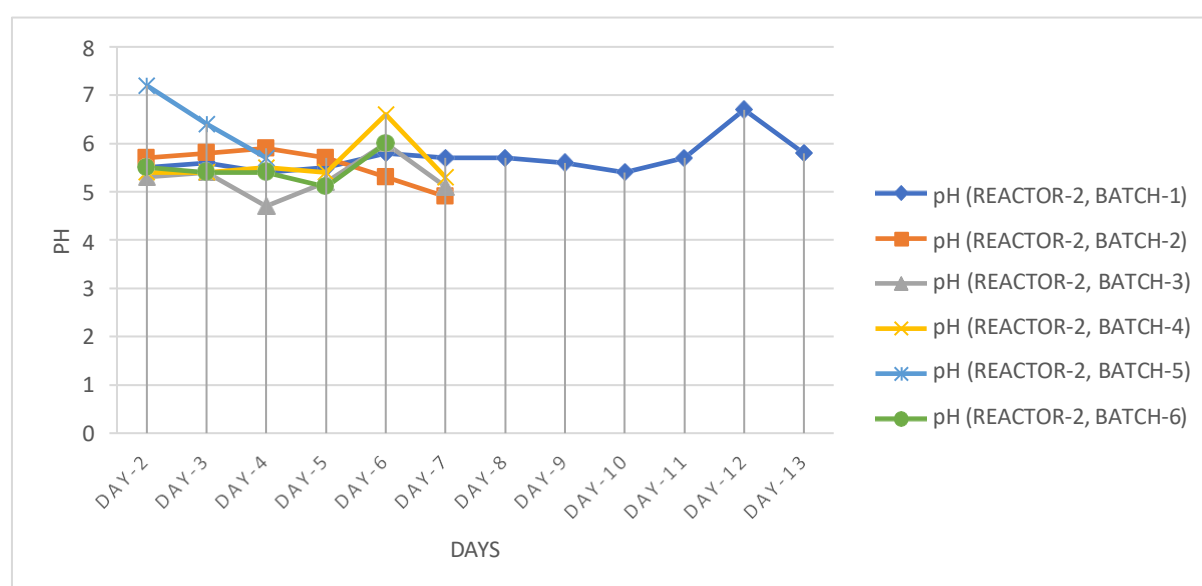


Figure 18: pH (Reactor-2) by Date

REACTOR-3 (All Batches)

In REACTOR-3, pH variations across batches were more dynamic, with some spikes that suggest significant biological changes. Batch-2, for instance, exhibited a sharp rise in pH from **5.8 to 7.0** by Day-6, followed by a gradual decrease, reflecting strong microbial activity or buffering effects. Similarly, Batch-5 started with an alkaline trend but decreased over time. Batch-4 showed noticeable fluctuations, possibly due to intermittent substrate availability or varying reaction rates. Notably, Batch-1 showed a late sharp increase (Day-12: **7.5**), which could be due to reduced acid production or external factors like aeration or feed composition changes.

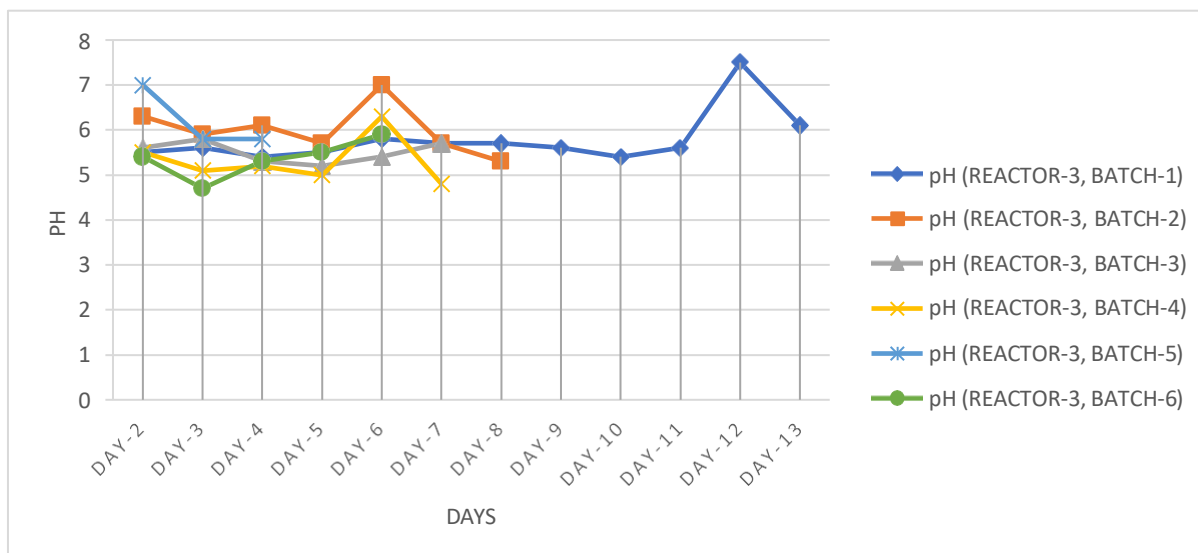


Figure 19: pH (Reactor-3) by Date

Table 2: Comparative pH Analysis of Reactors

Feature	Reactor-1	Reactor-2	Reactor-3
Overall Range	6.3 – 7.8	6.5 – 8.3	6.4 – 8.5
Batch-1	Stable (6.8 – 7.2)	Stable (6.8 – 7.2)	Stable (6.5 – 7.2)
Batch-2	Neutral and stable (~6.9 – 7.1)	Slight rise (~6.9 – 7.4)	Stable (~6.7 – 7.1)
Batch-3	Rises gradually (6.5 → 7.6)	Increases to ~7.8	Increases to ~8.2
Batch-4	Slight fluctuation (~7.0 – 7.5)	Rises to ~8.0	Rises to ~8.0
Batch-5	Stable (6.8 – 7.1)	Slight increase (~6.8 – 7.5)	Mild rise (~6.9 – 7.6)
Batch-6	Stable, mild increase (~6.8 – 7.1)	Sharp increase (6.8 → 8.3)	Highest increase (6.8 → 8.5)
Notable Observation	All batches near-neutral; Batch-3 rises	Alkaline shift in Batch-6 and Batch-4	Strongest alkaline trend in Batch-6

4.2. COD Variations

REACTOR-1 Observations

Overall COD Range: 0.5 to 14 mg/gm. **Batch-3** stands out with a sharp rise from **Day 2 (6 mg/gm)** to a peak at **Day 4 (14 mg/gm)**, then declines. **Other batches (1, 2, 4, 5, 6)** show a general decreasing trend, especially Batch-1, which stabilizes near **0.5 mg/gm**. **Day-7 onwards**, COD levels plateau at lower values across all batches.

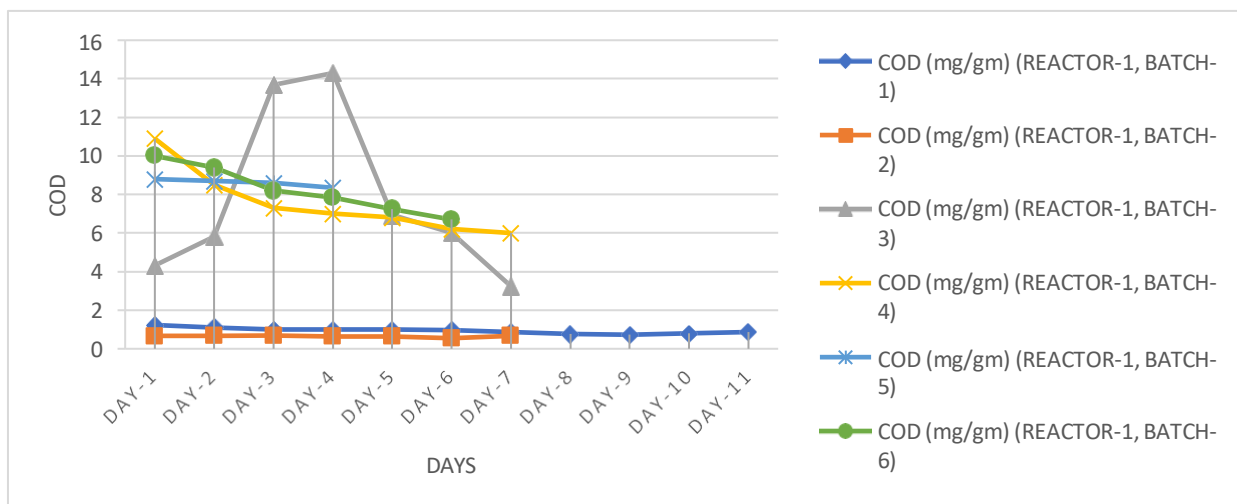


Figure 20: COD (Reactor-1) by Days

REACTOR-2 Observations

Overall COD Range: 0.5 to 17 mg/gm. **Batch-6** starts high (~17 mg/gm) and shows a steady decline to ~13 mg/gm by **Day-7**. **Batch-3** shows a similar mid-phase peak (13–14 mg/gm) and then declines. **Batches 1, 2, 4, 5** stabilize around **0.5 to 7 mg/gm**, with most dropping significantly after **Day-5**.

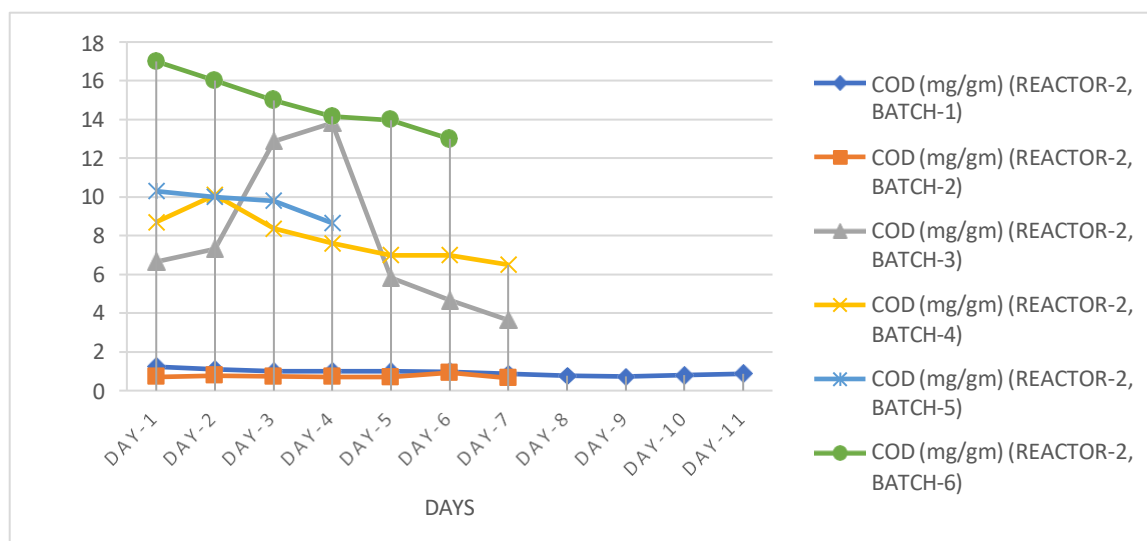


Figure 21: COD (Reactor-2) by Days

REACTOR-3 Observations

Overall COD Range: 0.5 to 28 mg/gm. **Batch-6** starts very high (~28 mg/gm) and exhibits a consistent linear decline to ~14 mg/gm by Day-7. **Batch-3** again shows a peak (~13 mg/gm on Day 4) before sharply dropping. **Batches 1, 2, 4, 5** perform similarly to Reactor-2 but with slightly more stable or lower values post Day-5.

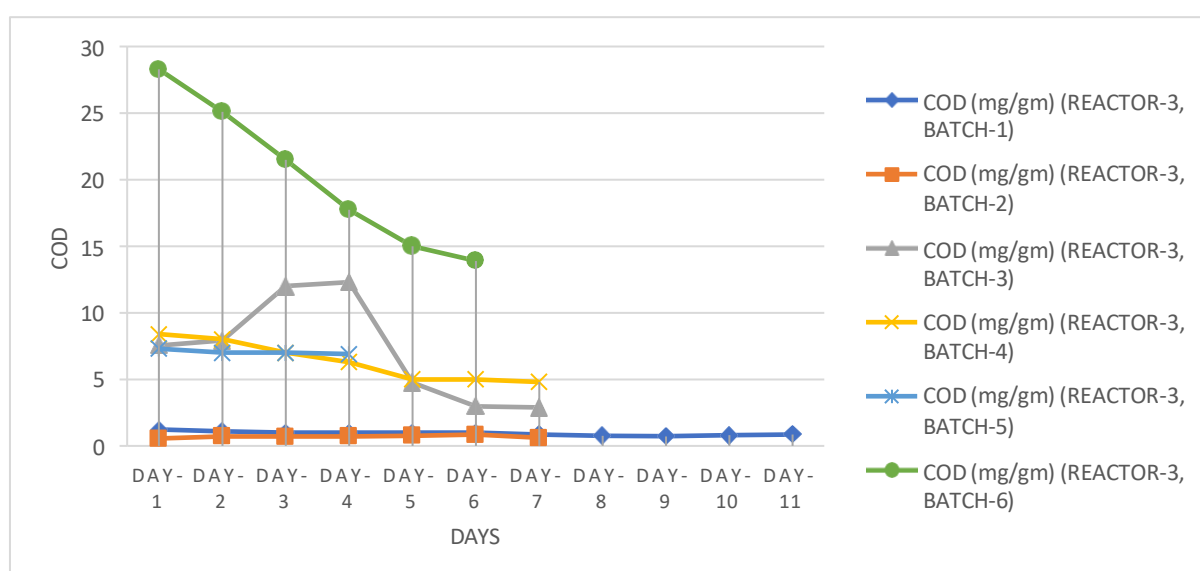


Figure 22: COD (Reactor-2) by Dates

Table 3: Key Comparative Insights- COD

Feature	Reactor-1	Reactor-2	Reactor-3
Max COD observed	~14 mg/gm (Batch-3)	~17 mg/gm (Batch-6)	~28 mg/gm (Batch-6)
COD Decline trend	Steady post-Day-4	Sharper post-Day-4	Strong linear decline in Batch-6
Best-performing batch	Batch-1 (early stability)	Batch-2 & 5 (flat low COD)	Batch-1 & 2 (flat COD)
Peak anomaly	Batch-3 (spike on Day-4)	Batch-3 (similar behavior)	Batch-6 (initially very high)

4.3. Temperature Variations

Temperature Variations in REACTOR-1 (All Batches)

The temperature variations in REACTOR-1 across the different batches displayed distinct patterns. Batch-1 maintained a stable temperature range between 24°C and 28°C, indicating a post-reaction or stabilization phase with minimal variation. Batch-2 fluctuated between 27°C and 32°C, with a peak on Day-4, suggesting moderate reaction dynamics. Batch-3 showed consistent temperatures ranging from 26°C to 31°C, reflecting stable microbial activity. Batch-4 exhibited the most dynamic behavior, with a peak at 34°C followed by a sharp drop to 27°C, indicating intense early reactions followed by cooling. Batch-5 was the most stable, with temperatures remaining between 31°C and 32°C, suggesting optimal reaction conditions. Batch-6 mirrored Batch-4, with an early peak at 34°C followed by a gradual cooling, signaling a strong initial reaction phase and subsequent stabilization. Overall, Batch-5 was the most stable, while Batch-4 showed the greatest fluctuations, and Batch-1 had the least variation, reflecting a more stabilized process.

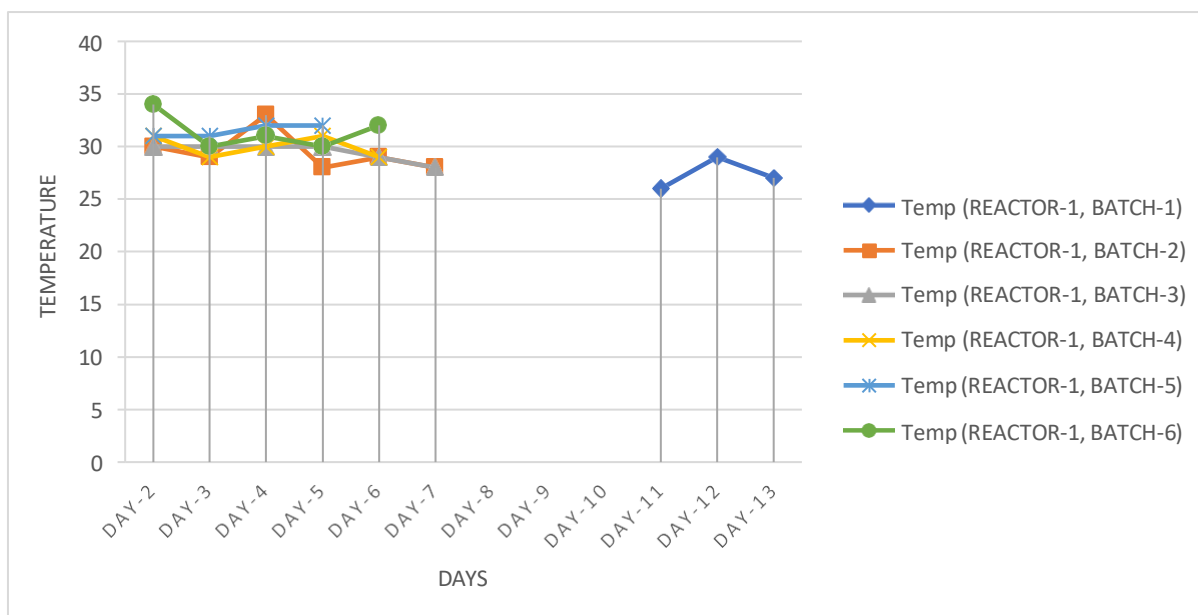


Figure 23: Temperature (Reactor-1) by Date

Temperature Variations in REACTOR-2 (All Batches)

Temperature readings in REACTOR-2 displayed a moderately stable pattern, though several batches exhibited notable deviations. Batch-5 began with a relatively high temperature of 33°C on Day-2, followed by a gradual decline, suggesting early exothermic activity likely linked to microbial growth. Batch-4 showed a short-lived temperature spike (Day-3: 34°C) before returning to more stable values, indicating a brief but intense reaction phase. In contrast, Batch-3 exhibited consistent mid-range temperatures (around 29–31°C), reflecting a controlled process with limited thermal disturbance. Batch-2 started cooler (28°C), but with a minor peak on Day-4, hinting at delayed microbial activity. Batch-1 remained inactive in the initial days, registering lower temperatures only during the later phase (Days 11–13), with values stabilizing between 24°C and 28°C, indicative of either a post-reaction or secondary stabilization process. Overall, Batch-4 and Batch-5 recorded the **widest thermal variations**, likely due to substrate concentration differences or metabolic surges.

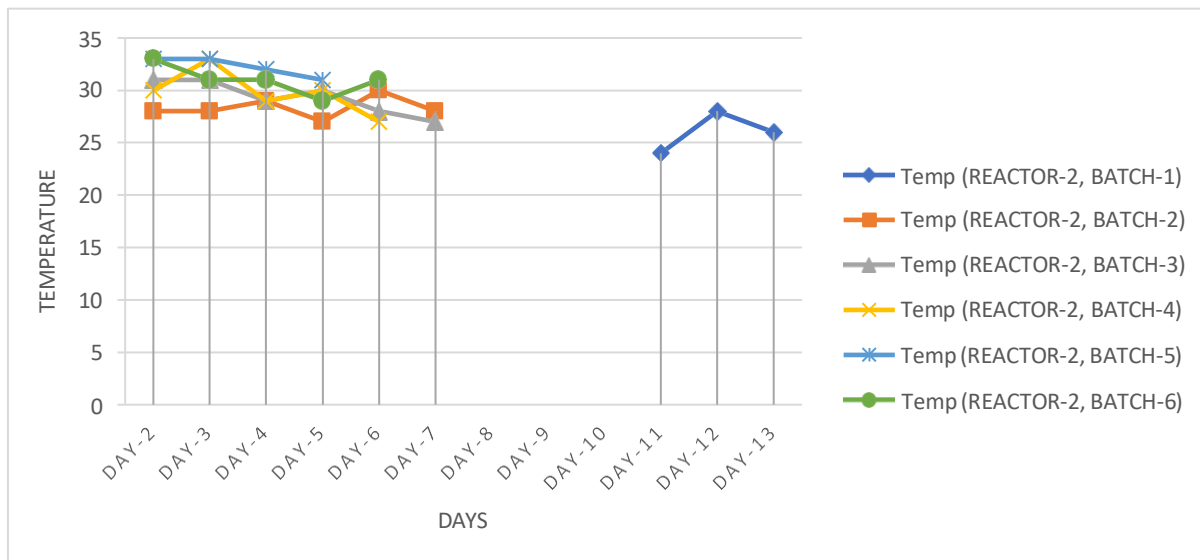


Figure 24: Temperature (reactor-2) by date

Temperature Variations in REACTOR-3 (All Batches)

Temperature behavior in REACTOR-3 was more dynamic, showing sharper peaks and drops across multiple batches, implying higher biological or chemical activity. Batch-2 demonstrated a sharp climb from 29°C to 32°C by Day-3, followed by a slow decline, possibly indicating rapid biomass buildup or metabolic heat generation. Batch-5 also started with a high temperature (34°C), which steadily dropped by Day-6, suggesting a rapid but contained reaction sequence. Notably, Batch-4 presented erratic thermal behavior, spiking at 33–34°C early in the cycle before dipping sharply—likely due to inconsistent substrate feeding or aeration. Batch-3 maintained a more balanced range (28–31°C), indicating efficient thermal regulation. A unique pattern emerged in Batch-1, where temperature remained low initially, but increased gradually during the later phase, with a peak at 28°C by Day-13. This delayed warming trend may suggest a lag in microbial activation or late substrate introduction. In summary, REACTOR-3 showed more pronounced **temperature spikes**, especially in Batches 2, 4, and 5, likely reflecting higher microbial turnover and variable system dynamics.

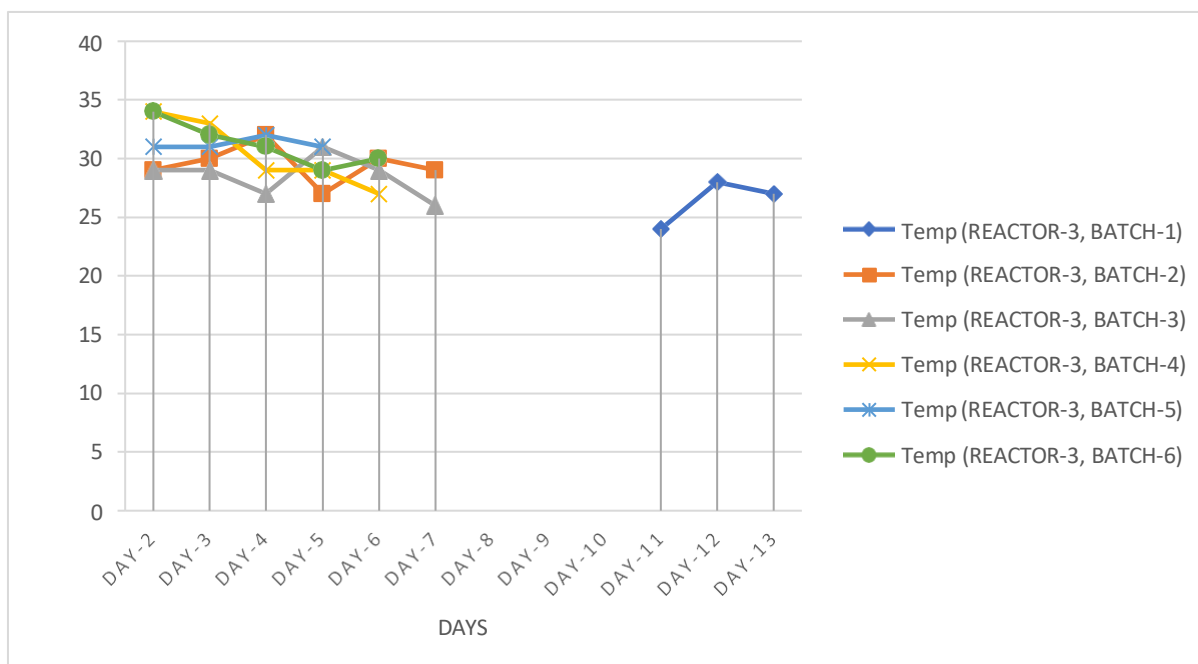


Figure 25: Temperature (reactor-3) by date

Table 4: Key insights of Temperature

Feature	Reactor-1	Reactor-2	Reactor-3
Max Temperature observed	34°C (Batch-4 & 6)	34°C (Batch-6)	34°C (Batch-6)
Temperature Decline trend	Gradual cooling after peak (Batch-4 & 6)	Fluctuations with peaks (Batch-5 & 4)	Sharp drop post-peak (Batch-6)
Most stable batch	Batch-5 (31°C–32°C)	Batch-1 (28°C–31°C)	Batch-1 (26°C–31°C)
Peak anomaly	Batch-4 (spike to 34°C)	Batch-4 (spike to 34°C)	Batch-6 (initially high at 34°C)

4.4. TKN (Total Kjendhal Nitrogen) Variations

REACTOR-1:

In REACTOR-1, TKN levels were relatively moderate, with most batches showing gradual reductions over time. Batch-3 exhibited the most significant fluctuation, peaking at **0.55 mg/L on Day-4**, indicating a temporary accumulation of organic nitrogen. Batches 1 and 2 displayed more stable trends, maintaining low TKN values throughout, with Batch-1 starting at **0.0467 mg/L** and gradually

declining to **0.035 mg/L** by Day-11. Batch-5 and Batch-6 showed slightly higher values initially but followed a steady downward trend. Overall, REACTOR-1 demonstrated **balanced nitrogen transformation**, with Batch-1 showing the most consistent performance.

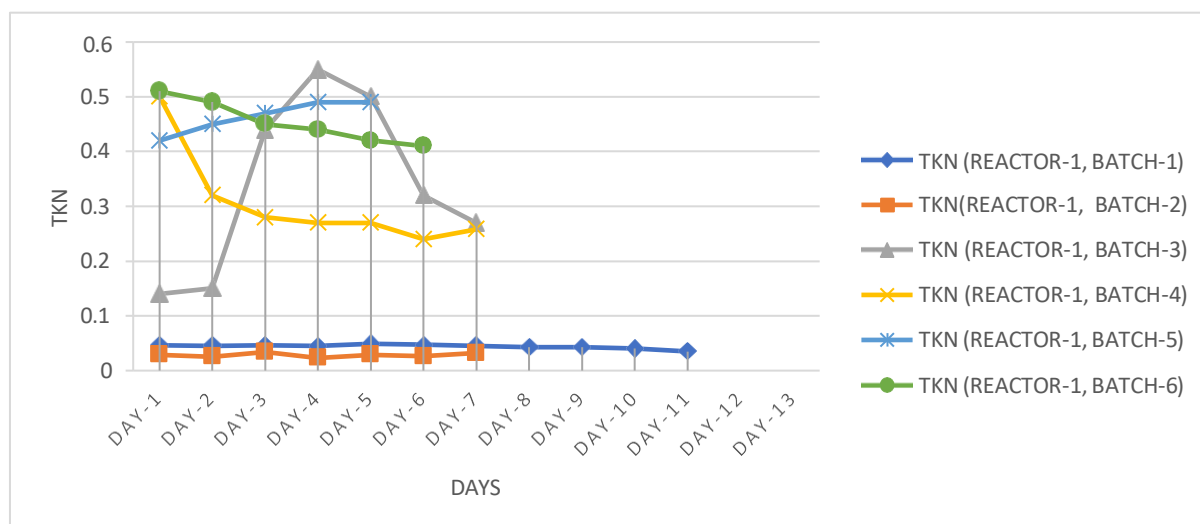


Figure 26: TKN (Reactor-1) By Date

REACTOR-2:

REACTOR-2 recorded comparatively **higher TKN values**, especially in **Batch-6**, which remained in the range of **0.85–0.94 mg/L** throughout the observed period. This suggests a persistent nitrogen load and slower degradation. Batch-3 also showed noticeable variation, peaking around **0.48 mg/L** before declining. Batch-2 had the lowest values, consistently hovering around **0.024–0.041 mg/L**, indicating efficient nitrogen reduction. Batch-1 followed a similar pattern of gradual decline. The variation across batches in REACTOR-2 indicates **moderate to high nitrogen activity**, with Batch-6 being the most nitrogen-rich and resistant to reduction.

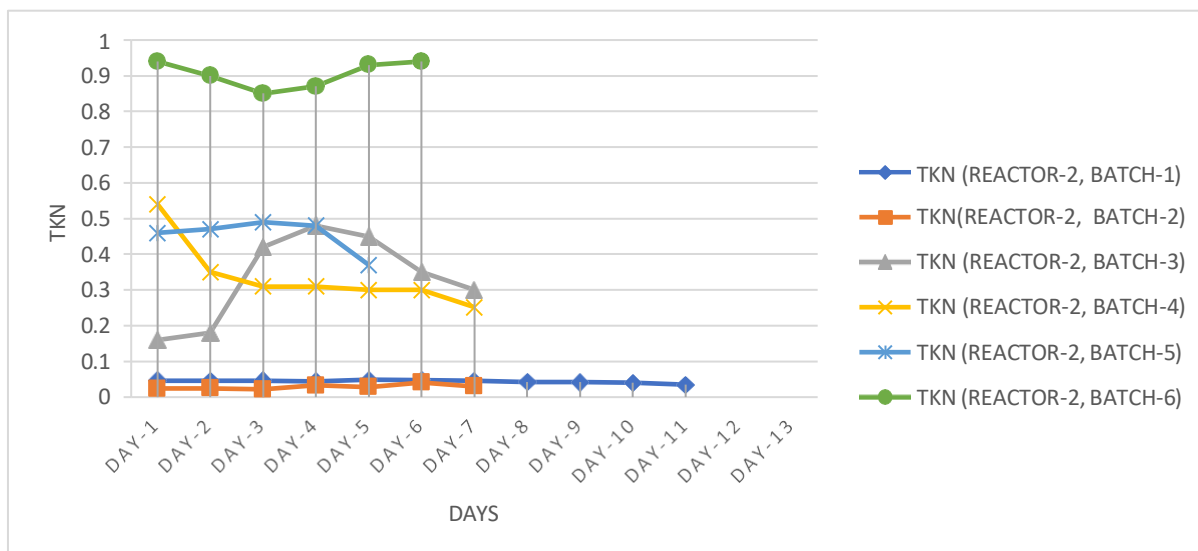


Figure 27: TKN (Reactor-2) by Date

REACTOR-3:

REACTOR-3 exhibited the widest range of TKN variation, with Batch-6 starting at 1.41 mg/L—the highest among all reactors—and gradually decreasing to 1.22 mg/L by Day-6, reflecting a high initial nitrogen load and sluggish degradation. Batch-3 also displayed dynamic behavior, with a sharp increase to 0.5 mg/L on Day-5 followed by a rapid drop, indicating strong microbial turnover or nitrogen cycling. Batch-1 remained stable with a slow and steady decline from 0.0467 mg/L to 0.035 mg/L, mirroring patterns seen in the other reactors. Among all setups, REACTOR-3 had the most intense nitrogen fluctuations, particularly in Batch-6, suggesting the need for further optimization in nitrogen removal.

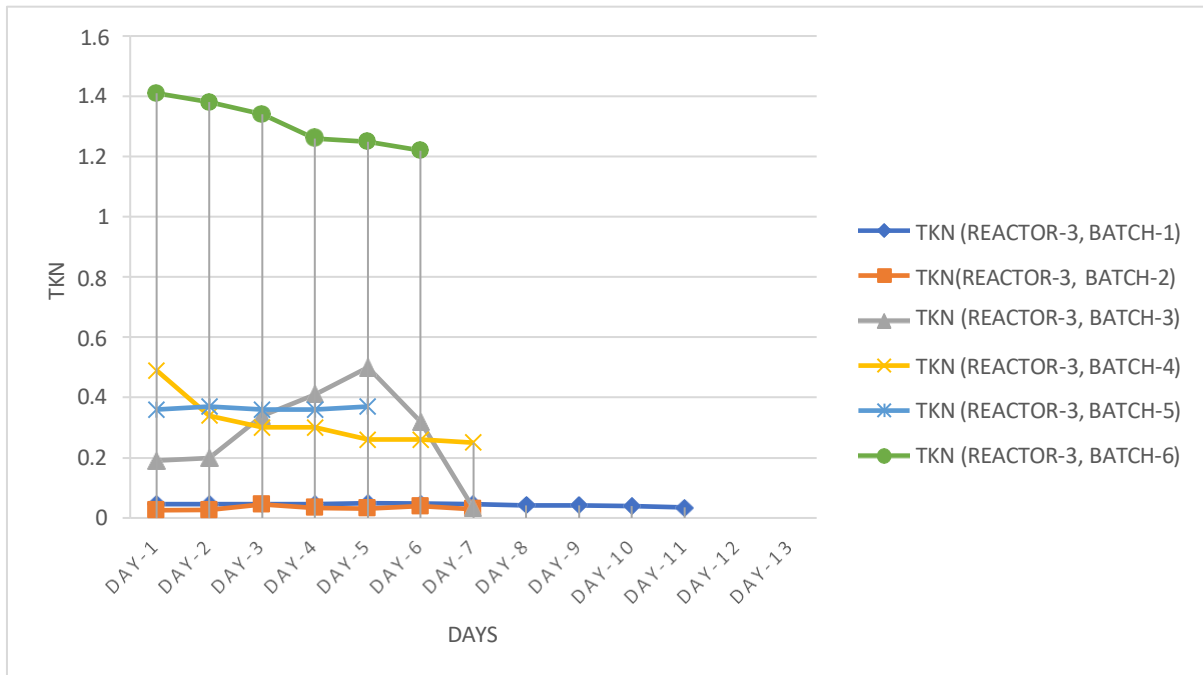


Figure 28: TKN (Reactor-3) by Date

Table 5 : Key Comparative Insights – TKN Across All Reactors

Feature	REACTOR-1	REACTOR-2	REACTOR-3
Max TKN observed	0.55 (Batch-3, Day-4)	0.94 (Batch-6, Days 1 & 6)	1.41 (Batch-6, Day-1)
Min TKN observed	0.023 (Batch-2, Day-4)	0.022 (Batch-2, Day-3)	0.0256 (Batch-2, Day-1)
Most stable batch	Batch-1 (gradual decline)	Batch-1 & 2 (low, steady TKN)	Batch-1 & 5 (least fluctuation)
Most dynamic batch	Batch-3 (sharp peak at Day-4)	Batch-3 & 6 (active variation with high values)	Batch-3 (sharp rise-fall)
Consistently high TKN	Batch-5 & 6 (moderately high)	Batch-6 (0.85–0.94 range)	Batch-6 (1.22–1.41 range)
TKN decline pattern	Mostly declining post-Day-4	Gradual decline in Batches 3–6	Sharp fall post Day-5 in Batch-3, slow in Batch-6

4.5. C/N RATIO VARIATIONS

Reactor-1 Analysis:

In Reactor-1, the C/N ratios for various batches exhibit different trends over the 7-day period. For Batch-2, the ratio fluctuates from 23.103 on Day-1 to a peak of 30.22 on Day-2 before dropping to 20.29 on Day-3 and then reaching 22.068 on Day-5. Batch-3 starts with a C/N ratio of 30.85 on Day-1, then increases slightly to 38.8 on Day-2, before decreasing to 30.5 on Day-3. The ratio further drops to 13.88 on Day-5, reaching its lowest value. Batch-4 shows a gradual decline from 28.5 on Day-1 to 25.18 on Day-5, with the ratio slightly varying between 18 and 26 during the initial days. Batch-5 starts at 20.95 on Day-1, then decreases to 19.06 on Day-2, with a consistent decline reaching 17 on Day-5. Batch-6's ratio stays relatively lower compared to others, starting at 19.5 on Day-1 and gradually decreasing to 16.3 on Day-6, with a slight dip to 16.25 on Day-7.

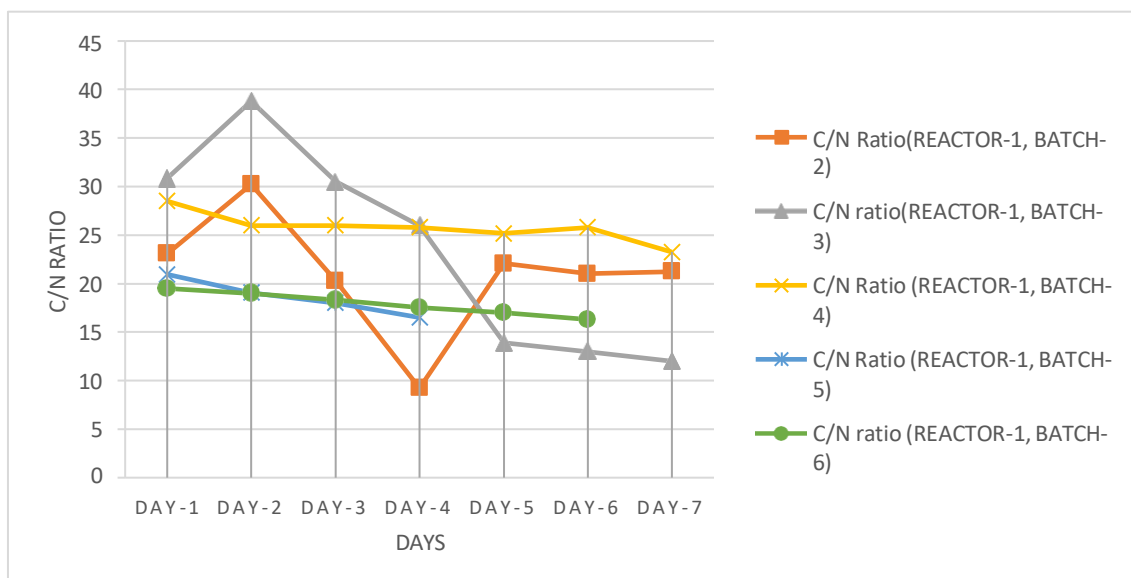


Figure 29:C/N ratio (reactor-1) by date

Reactor-2 Analysis:

In Reactor-2, the C/N ratios display varying patterns across the batches. Batch-2 begins at 29.58 on Day-1, rises slightly to 31.42 on Day-2, and maintains a stable

range between 25 and 33 during the subsequent days. Batch-3 starts higher, with a ratio of 41.68 on Day-1, reaching its peak at 40.66 on Day-2, before dropping to 34.9 on Day-3 and further declining to 12.93 by Day-5. The trend in Batch-4 begins at 26.2 on Day-1, remains steady at around 28.3 on Day-2, and decreases consistently throughout the week. Batch-5's C/N ratio starts at 22.39 on Day-1, steadily declines to 20.67 by Day-2, and reaches a lower value of 12.93 on Day-5. Batch-6's ratio is the lowest from the beginning, starting at 18 on Day-1 and declining gradually to 14.2 by Day-6.

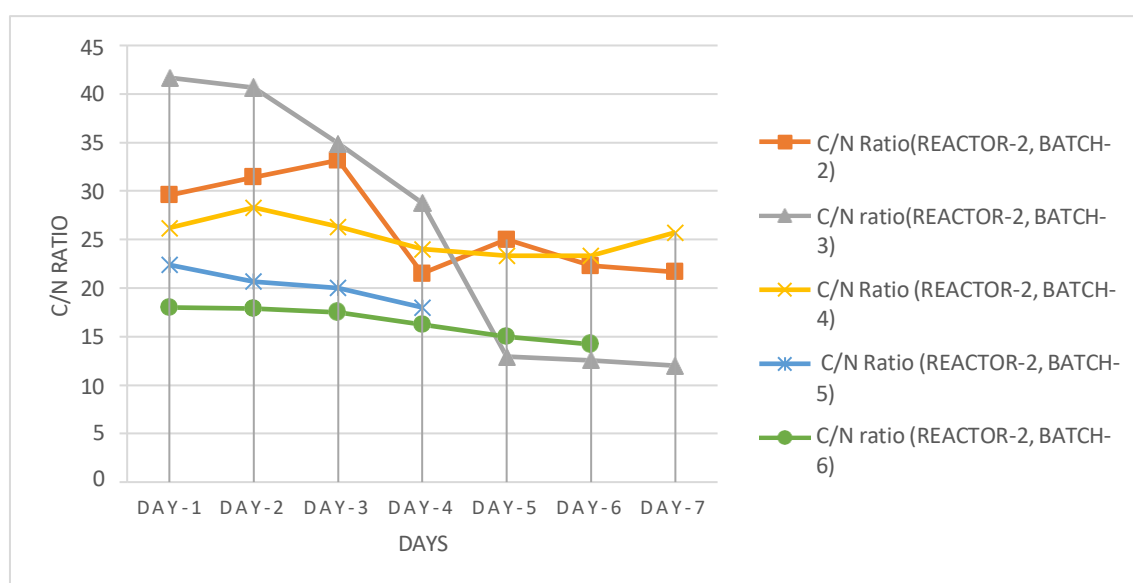


Figure 30: C/N ratio (reactor-2) by date

Reactor-3 Analysis:

In Reactor-3, Batch-2 begins with a C/N ratio of 21.71 on Day-1, rising to 26.79 on Day-2, before dropping to 21.17 by Day-3, and maintaining this value until Day-5. Batch-3 starts at 39.73 on Day-1, decreases slightly to 39.3 on Day-2, and then sees a consistent drop to 33.2 by Day-3, reaching 30 by Day-4. Batch-4's ratio starts at 23.8 on Day-1 and decreases gradually to 22.8 on Day-2 before dipping to around 19.23 by Day-6. Batch-5 starts at 20.27 on Day-1, with a slight decline to 19.75 on Day-2, and further drops to 19.15 by Day-4, remaining steady

at this level. Batch-6 follows a similar downward trend, with the ratio starting at 20 on Day-1 and dropping steadily to 10.8 by Day-6.

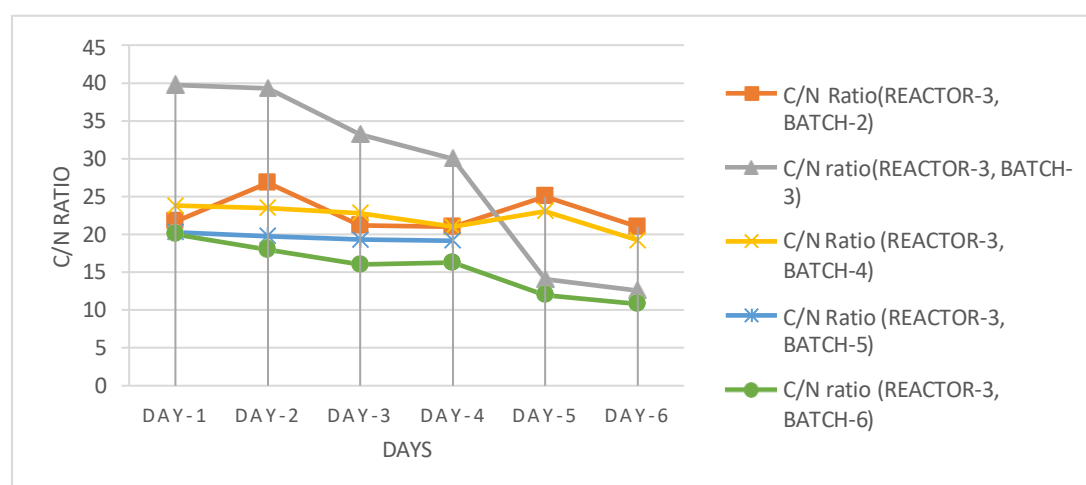


Figure 31: C/N ratio (reactor-3) by date

Table 6: Key insights of C/N ratio

Insight	Reactor-1	Reactor-2	Reactor-3
Fluctuation Over Time	Fluctuates significantly, especially in some batches (e.g., Batch-2)	Fluctuates, with remains relatively steady in some batches (e.g., Batch-3)	Fluctuates but remains relatively steady in some batches (e.g., Batch-2)
Batch-Specific Behavior	Batch-2 is more stable; Batch-3 has high variation	Batch-3 has high variation; Batch-6 is lowest	Batch-2 is stable; Batch-3 varies more
Trend of C/N Ratio Over Time	Decreases over time, especially in Batch-6	Decreases over time in all batches	Generally, decreases, with Batch-5 showing significant drops
Lower Ratios Over Time	C/N Observed in most batches, indicating carbon consumption	Observed in most batches, indicating carbon consumption	Observed in most batches, showing active degradation
Reactor-Specific Behavior	More varied across batches, Batch-3 stays high	Generally higher ratios than others, especially in Batch-3	Consistent ratios, especially in Batch-2
Nutrient Imbalance Risk	High variability may indicate imbalance	Higher ratios may suggest a carbon-rich environment	Lower ratios indicate nitrogen limitation

Insight	Reactor-1	Reactor-2	Reactor-3
Batch-to-Batch Variation	Significant differences, especially Batch-3	High variability across all batches	Moderate variation between batches
Process Optimization Potential	Requires adjustments based on batch needs	Potential for optimizing carbon/nitrogen balance	Stable conditions suggest potential for controlled adjustments

4.6. Moisture Content Analysis

Reactor-1 Analysis:

In Reactor-1, moisture content varies noticeably across batches. **Batch-1** has data only from Day-11 onwards, showing a significant drop from 73% to 52%, then stabilizing at 53% on Day-13. This late-stage drying suggests prolonged moisture loss. **Batch-2** maintains a high and relatively stable MC between 74% and 84%, indicating effective moisture retention and likely optimal microbial activity. **Batch-3** shows an unusual trend: starting at 74%, it drops sharply to 39.2% by Day-5, then rebounds to 74–80% by Day-7—possibly due to rehydration or inconsistent mixing. **Batch-4** sees a steady decline from 71% to around 52% by Day-5, suggesting drying, though data beyond Day-5 is unavailable. **Batch-5** decreases steadily from 62.8% to 45.4%, reflecting ongoing water loss. Similarly, **Batch-6** starts with a low 49.3% and gradually drops to 45.9% by Day-6, suggesting a need for improved moisture control to sustain microbial processes.

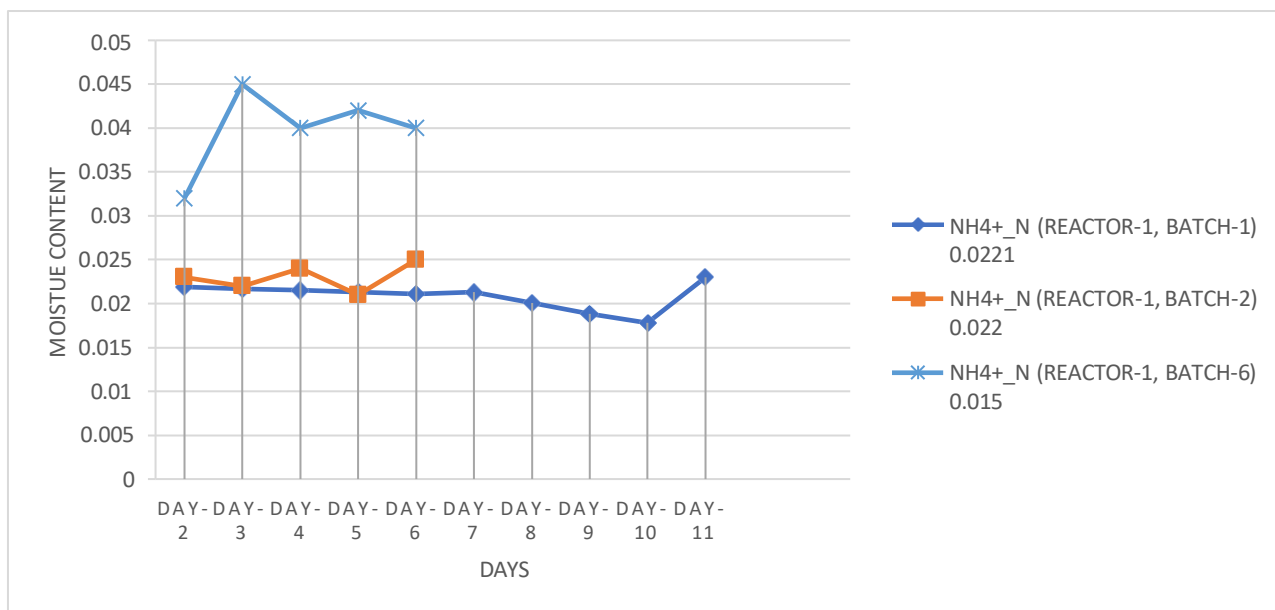
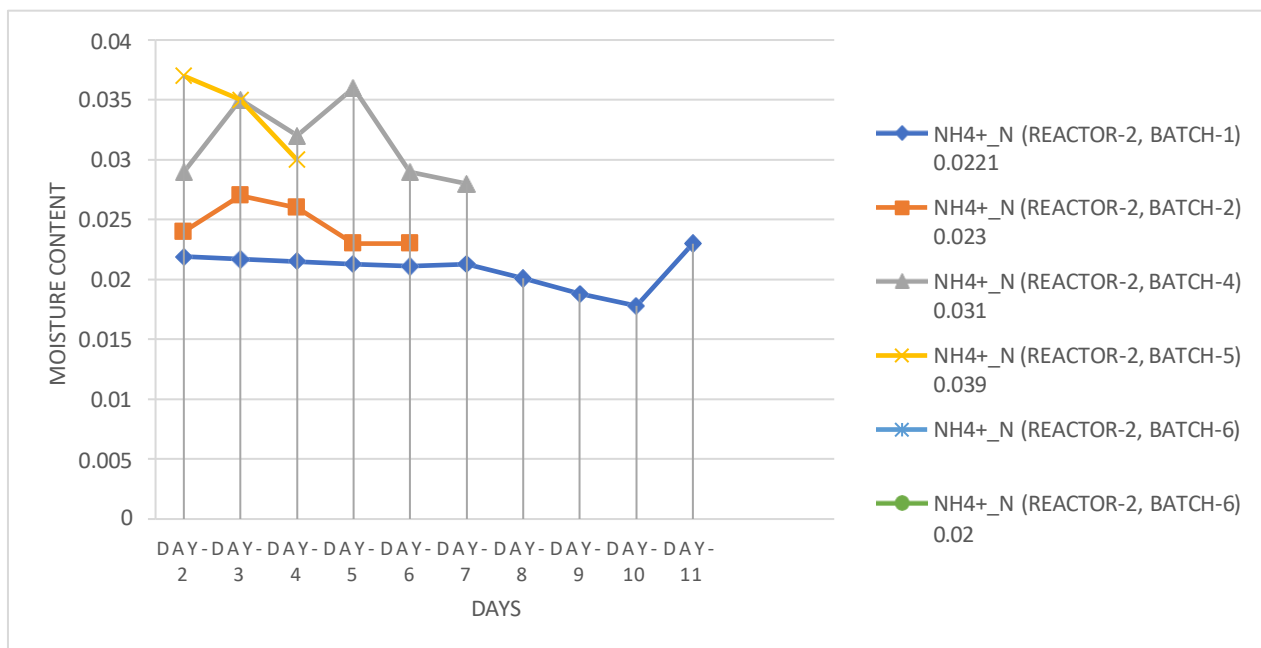


Figure 32: Moisture Content (Reactor-1) by Date

Reactor-2 Analysis:

In Reactor-2, **Batch-1** also has sparse data, showing a decrease from 58% (Day-11) to 49% (Day-12) and a slight recovery to 52% (Day-13), indicating moisture loss over time. **Batch-2** displays excellent stability, starting at 84% and staying consistently above 74%—a favorable condition for biodegradation. **Batch-3** follows a similar trend, fluctuating slightly between 70–79%, indicating good moisture management. **Batch-4**, however, drops quickly from 71% to 44.5% in just two days, then hovers in the mid-40s, showing early drying. **Batch-5** declines from 72% to 44.5% by Day-4, then stabilizes slightly above 49%, again suggesting drying during the initial phase. **Batch-6** shows a steady decline from 61.1% to 56% over six days, maintaining moderate moisture but trending downward.



Reactor-3 Analysis:

For Reactor-3, **Batch-1** follows the same trend as other reactors, with data from Day-11 onward showing a drop from 67% to 50%, then a slight rise to 55% by Day-13, indicating late-stage drying. **Batch-2** is consistent and moist throughout, with values around 70–78%, which is ideal for microbial activity. **Batch-3** presents a sharp decline from 74% to 32.9% by Day-5, then recovers to 71% by Day-7—indicative of a possible moisture addition after drying. **Batch-4** fluctuates slightly, increasing to 71% initially but dropping to the 50s by Day-5, showing gradual moisture loss. **Batch-5** consistently drops from 59% to 40.7% over six days, reflecting steady drying. **Batch-6** starts strong at 59.3%, climbs to 65.2%, but drops sharply to 40.7% by Day-6—indicating potential issues with moisture retention in the latter half.

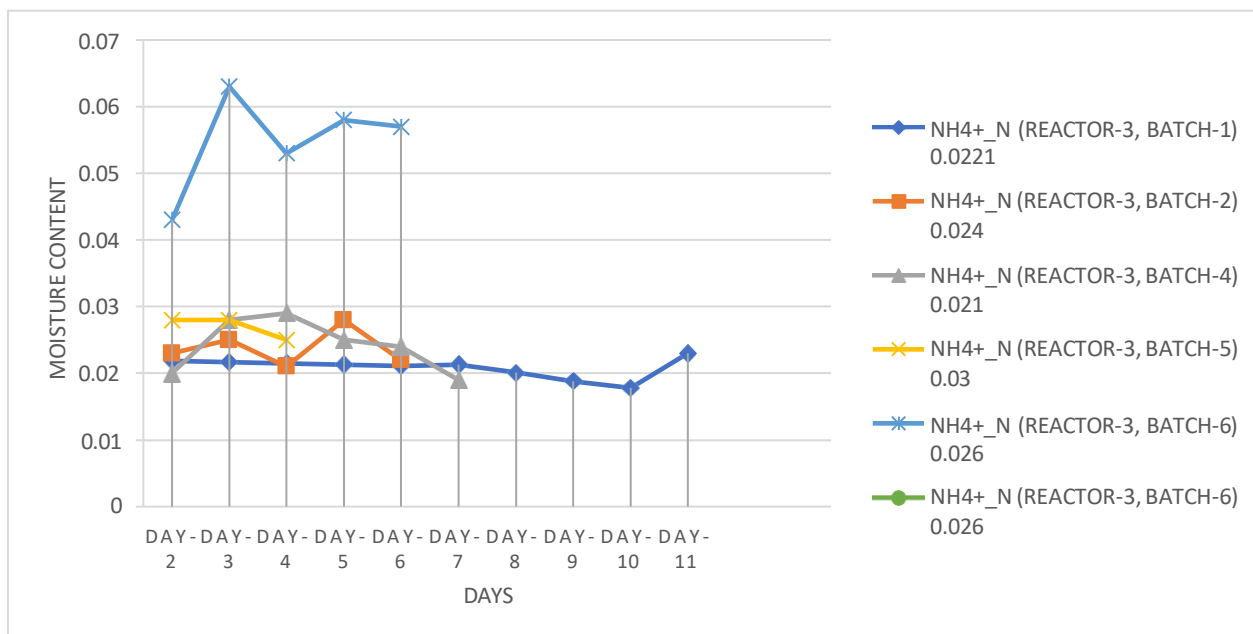


Figure 33: Moisture Content (Reactor-3) by Date

Table 7: Key Insights of Moisture Content

Parameter	Reactor-1	Reactor-2	Reactor-3
Moisture Stability	Batch-2 and Batch-3 show high and consistent MC	Batch-2 and Batch-3 are most stable	Batch-2 most stable; Batch-3 shows sharp fluctuations
Sudden Drops	Batch-3 (Day-5), Batch-5	Batch-4 and Batch-5 (early days)	Batch-3 (Day-5), Batch-6 (Day-6)
Low MC Risk	Batch-5, Batch-6	Batch-4, Batch-5	Batch-5, Batch-6
Late Stage Drying	Batch-1 (post Day-10)	Batch-1 (post Day-10)	Batch-1 (post Day-10)

4.7. NH₄⁺-N (Ammonium) Variations

Reactor-1 Analysis:

In **Reactor-1**, NH₄⁺-N levels start low across all batches, with slight day-to-day variations. **Batch-1** begins at 0.0221 mg/L on Day-1 and gradually declines to 0.0178 mg/L by Day-10, indicating steady ammonification followed by possible nitrification or volatilization. **Batch-2** fluctuates slightly between 0.021 and

0.025 mg/L during Days 1–6, suggesting stable ammonium levels with minor microbial shifts. **Batch-4** shows an increasing trend from 0.028 to 0.032 mg/L by Day-4, possibly due to active organic nitrogen breakdown, then stabilizes near 0.028 mg/L. **Batch-5** starts at a higher 0.04 mg/L, peaks at 0.05 mg/L by Day-4, indicating significant ammonification, though data beyond Day-4 is missing. **Batch-6** rises from 0.015 mg/L to a high of 0.045 mg/L on Day-3, maintaining a high level (around 0.04–0.042) till Day-6, reflecting intense microbial activity and ammonium accumulation.

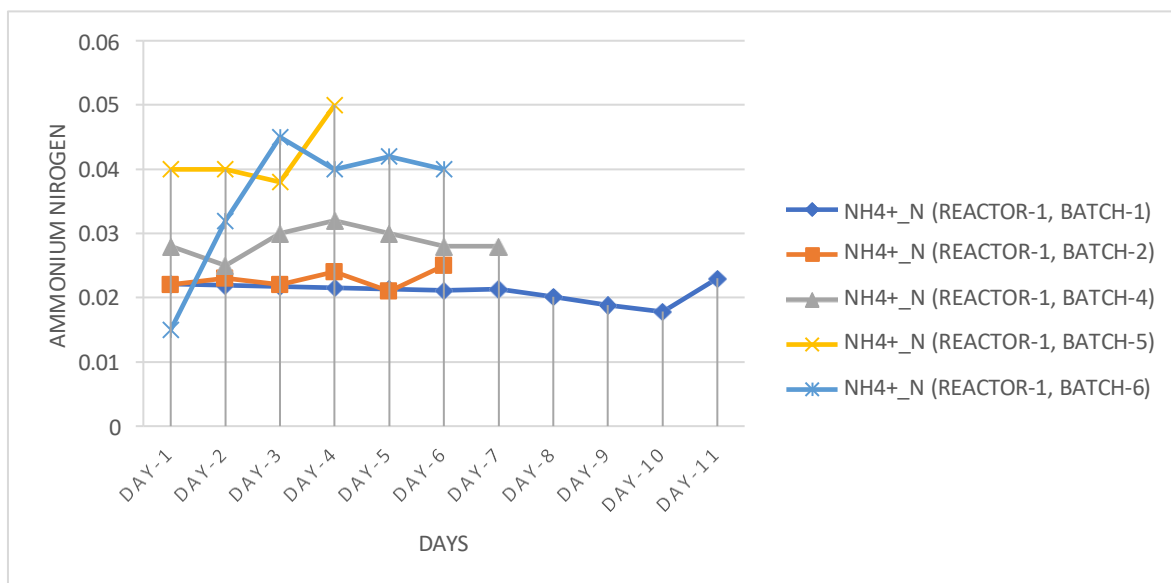


Figure 34: Ammonium Nitrogen (Reactor-1) by Date

Reactor-2 Analysis:

Reactor-2 mirrors many of Reactor-1's trends but with slightly elevated $\text{NH}_4^+\text{-N}$ levels. **Batch-1** follows the same declining path from 0.0221 to 0.0178 mg/L across Days 1–10, signifying nitrogen mineralization over time. **Batch-2** ranges between 0.023 and 0.027 mg/L during Days 1–4, reflecting moderate ammonification, with a decline afterward. **Batch-4** exhibits a sharper rise from 0.031 mg/L on Day-1 to 0.036 mg/L by Day-5, then a slight drop, likely due to initial organic decomposition followed by stabilization. **Batch-5** sees a decrease from 0.039 to 0.03 mg/L over four days, indicating $\text{NH}_4^+\text{-N}$ uptake or

transformation. **Batch-6** demonstrates a noticeable surge from 0.02 to 0.055 mg/L by Day-6—one of the highest among all reactors—suggesting strong ammonification and perhaps insufficient nitrification.

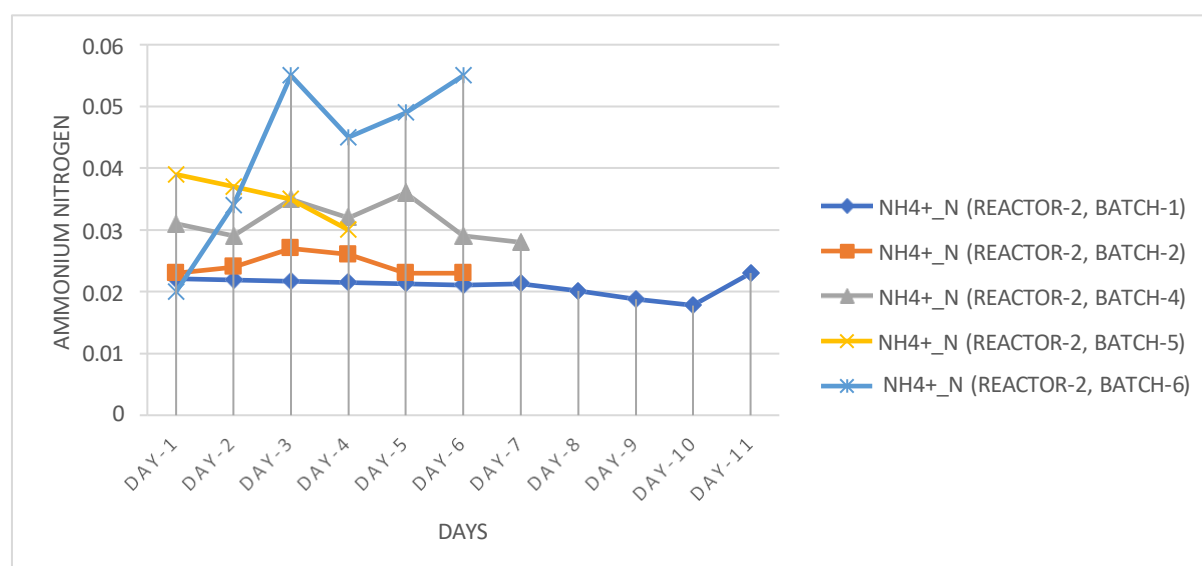


Figure 35: Ammonium Nitrogen (Reactor-2) by Date

Reactor-3 Analysis:

In **Reactor-3**, the trends are more dynamic. **Batch-1** follows a gradual decline similar to Reactors 1 and 2, moving from 0.0221 to 0.0178 mg/L. **Batch-2**, however, shows more fluctuation—initially at 0.024 mg/L, it drops slightly, spikes at 0.028 mg/L on Day-5, then drops again—indicative of variable ammonium release and microbial activity. **Batch-4** begins low at 0.021 mg/L but rises to 0.029 mg/L by Day-4, then slightly decreases, likely due to organic matter breakdown followed by microbial assimilation. **Batch-5** exhibits a gradual drop from 0.03 to 0.025 mg/L by Day-4, suggesting consistent NH_4^+ consumption or conversion. **Batch-6** displays a significant rise—from 0.026 to 0.063 mg/L by Day-3—then remains elevated around 0.057–0.058 mg/L, highlighting intense ammonification and accumulation with limited transformation.

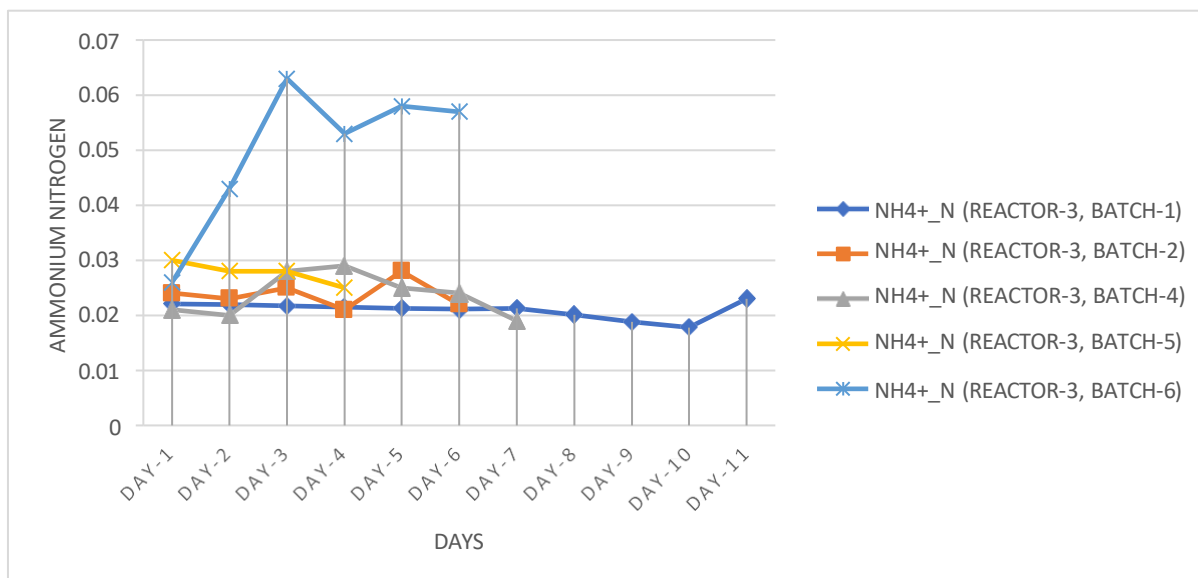


Figure 36: Ammonium Nitrogen (Reactor-3) by Date

Table 8: Key Insights of Ammonium

Parameter	Reactor-1	Reactor-2	Reactor-3
General Trend	Gradual increase in $\text{NH}_4^+\text{-N}$ in most batches (esp. Batch-5 & Batch-6); some show stable or declining trend	More pronounced ammonification, especially in Batch-6; Batch-4 shows active N release	High $\text{NH}_4^+\text{-N}$ peak in Batch-6; generally rising trends with dynamic behavior
Peak $\text{NH}_4^+\text{-N}$ Batch	Batch-6 (0.045 mg/L on Day-3)	Batch-6 (0.055 mg/L on Day-6)	Batch-6 (0.063 mg/L on Day-3) – highest among all reactors
Lowest $\text{NH}_4^+\text{-N}$ Batch	Batch-1 – gradual decline to 0.0178 mg/L	Batch-1 – similar decline pattern	Batch-1 – consistent decrease to 0.0178 mg/L
Stability Observed	Batch-2 – minor fluctuations (0.022 to 0.025)	Batch-2 – steady rise then flat	Batch-2 – fluctuation within 0.021–0.028 range
Sharp Increase	Batch-6 and Batch-5 – strong ammonification phase	Batch-6 – clear ammonification peak	Batch-6 – most aggressive ammonification
Microbial Activity Indicator	Moderate across batches, with delayed	High in Batch-4 and 6 – reflects microbial nitrogen conversion	Highest microbial conversion seen in Batch-6

Parameter	Reactor-1	Reactor-2	Reactor-3
	ammonification in Batch-4 and 5		

5. Conclusion

This study validates the feasibility of short-term aerobic composting as a practical and efficient solution for managing biodegradable components of municipal solid waste (MSW), particularly in urban and institutional settings. The composting process, conducted using a simple and low-cost pit method, effectively reduced organic waste volume and produced nutrient-rich compost suitable for soil application [1]. The observed temperature and pH trends were consistent with expected aerobic decomposition patterns, indicating robust microbial activity throughout the process [5]. Final compost quality parameters, including texture, odor, and maturity indicators, remained within acceptable limits, further supporting its agricultural applicability. The short-duration composting approach achieved maturity in approximately two weeks—significantly faster than conventional methods addressing critical space and time constraints commonly faced in urban environments [7][6]. Overall, this method presents a scalable, sustainable, and decentralized model for organic waste management that aligns with national waste reduction goals and global sustainability frameworks, including the United Nations Sustainable Development Goals [8][5].

6. REFERENCES

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